

Aditya Rao

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EDUCATION

McMaster University

Hamilton, ON

B.Eng in Software Engineering (CO-OP)

Sept 2023 – Expected Apr 2028

- 4.00/4.00 CGPA - Dean's Honour List
- Relevant Coursework: Intro to Software Design, Concurrent Systems Design, Data Structures and Algorithms, Software Requirements, Signals and Systems, Computer Architecture, Linear Optimization, Databases

EXPERIENCE

Software Developer Intern

May 2025 – Aug 2026

IBM

Markham, ON

- Developed an AI tool for Liberty build failures, automating root-cause log classification to save ~4 hours per break.
- Modernized legacy tools (Ruby on Rails, WAS 9 EJB) for 400+ developers into Java 21 microservices (Spring Boot, Next.js, Jakarta EE 11), cutting publish times from ~2 hours to <1 min via Kafka messaging.
- Engineered a work-distribution analytics dashboard in React and Quarkus, centralizing task tracking across CVEs to streamline project reporting and eliminate manual spreadsheets.
- Refactored 20+ legacy FAT test suites into modular JUnit frameworks to improve WebSphere CI/CD reliability.
- Contributed bug fixes and features to open-source OpenLiberty and WebSphere Application Server, resolving multi-platform runtime failures across Linux, macOS, Windows, and z/OS.

Full Stack Developer Intern

May 2024 – Aug 2024

Medium AI

Hamilton, ON

- Engineered real-time doctor-patient voice-based transcription with gRPC and Firebase, achieving sub-500ms streaming latency by load balancing OpenAI Whisper across self-hosted and managed instances.
- Integrated AI diagnostic hooks to analyze transcripts, highlighting clinical insights for attending physicians.
- Constructed responsive React and TypeScript UI for real-time transcription playback and physician dashboards.

PROJECTS

BYoAI Deployment Platform | 🌐 | Go, TypeScript, Docker, MCP, OpenTelemetry

- Engineered an AI agent orchestration platform in Go and TypeScript with tool-calling and observability.
- Containerized an execution engine with Docker for dynamic worker allocation, security isolation, and pluggable storage.
- Designed a multi-provider LLM abstraction supporting frontier and self-hosted models.

WebCraft | 🌐 | C++, CMake, Vcpkg, Asynchronous Programming, Github Actions

- Authored an asynchronous library in C++23 featuring coroutine task management, timers and scheduling, concurrent streams, and high-performance file and network I/O modules to simplify cross-platform development.
- Automated testing and release workflows using gtest (130+ tests), GitHub Actions CI/CD (with runners on Linux GCC, Linux Clang, Windows, and MacOS), and published a custom vcpkg port for standardized cross-platform distribution.

Resume Builder | 🌐 | Java, Spring Boot, Next.js, Typescript, Python, FastAPI, LangChain, Docker

- Built an AI resume builder with Next.js, Spring Boot, and FastAPI for real-time editing and version control.
- Integrated LangChain to automate ATS keyword optimization, intelligent scoring, and template tailoring.
- Implemented UI debouncing and reactive state management to eliminate render flickering for live resume previews.

LEADERSHIP

Sys-Admin and Fullstack Developer Executive

Nov 2024 – Present

IEEE McMaster Student Branch

Hamilton, ON

- Set up an RPi 3B+ cluster handling up to 100 concurrent clients with <50ms response latency.
- Provisioned CI/CD pipelines to build and deploy container images to a Kubernetes cluster.
- Developed the IEEE branch website using Next.js with GitHub Projects/Issues as an admin CMS.

TECHNICAL SKILLS

Languages: Java, JavaScript/TypeScript, Python, C#, C++, Golang, SQL, C, Verilog

Frameworks: Spring Boot, Jakarta EE, MicroProfile, Hibernate, Express, Next.js, FastAPI, Flask

Developer Tools: AWS, GCP, Git, Docker, Kafka, OpenShift, Kubernetes, Redis, Infinispan

Libraries: Pandas, TensorFlow, OpenCV, SciKit Learn, Redux, React